

The background features a complex network of thin, light gray lines forming a grid-like structure. Scattered throughout are numerous squares of various sizes and colors, including black, white, gray, yellow, green, blue, orange, and purple. Some squares are solid, while others are outlined. The overall aesthetic is modern and technological.

AR/VR Workshop

Landing on mars

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Outline

Geometric objects besides A-frame standard objects

Windows: Paint3 D

Linux, MacOS, Windows: Blender

Using a glb file

Ball Drop Code Explanation

In the component's schema

startY — how high up the ball starts (default: 5)

groundY — the height of the floor, so the ball knows where to stop (default: 0.5)

gravity — acceleration due to gravity (default: $\sim 9.8 \text{ m/s}^2$ at sea level)

bounce — speed the ball keeps after each bounce (default: 0.6, meaning it keeps 60% of its speed)

Init function

Init is run only once on creation of the A-frame component

1. Init sets the ball's starting speed to 0 (not moving yet) and places it at the starting height.
2. It sets up a click listener on the ball — so if you click it, it jumps back to the starting height and its speed resets to 0. This is what lets you drop it again after it lands
3. Ensure to remove the event listener in the remove function

Tick function

Every time the frame is drawn (redrawn)

1. **Update the ball's speed.** It takes the current downward speed and adds a bit more to it, based on gravity. Since this happens dozens of times per second, the ball doesn't just fall at a constant rate — it keeps speeding up the longer it falls, exactly like a real object would.
2. **Move the ball.** Now that we know how fast the ball is moving this frame, we use that speed to actually change its position — moving it down a little bit.
3. **Check if the ball hit the ground.** If the ball's current height is at or below the floor level, two things happen:
 - a) the ball gets snapped exactly onto the floor (so it doesn't look like it's sinking into it), and
 - b) its speed gets flipped from falling to bouncing — but reduced, using that “bounce” setting. So if it was falling fast, it now bounces back up, but slower than it was falling.
 - c) There's a check that once the bounce speed gets really small, it stops.

GLB loader – down in WebGL

GLB or GLTF

```
<head>
  <script src="https://aframe.io/releases/1.3.0/aframe.min.js"></script>
  <script src="https://cdn.jsdelivr.net/gh/donmccurdy/aframe-extras@v6.1.1/dist/aframe-extras.min.js">
</script>
</head>
<body>
  <script>
    document.getElementById("spaceShip").setAttribute("animation-mixer", "clip:swimming");
    const timedelay = setTimeout(delayFunction, 2000);
    function delayFunction() {
      document.getElementById("spaceShip").setAttribute("animation-mixer", "clip:bite");
    }
  </script>
```

Space ship and green cube

```
<a-scene>
  <a-entity id="spaceShip"
    gltf-model="./SpaceShip.glb"
    position=".5 0.5 -5" scale="2 2 2"></a-entity>
    <a-box position="0 -2 -5" color="green"> </a-box>
</a-scene>
```

Simulating some physics

Thanks

Ani (Anirudh) Magesh helped with the ball drop code and the ball-drop write up here

